

A Fine-Grained Study of Long-Tail Entities in News Articles

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Abstract

The fast-paced nature of modern news continuously introduces new entities—such as emerging public figures, local organizations, and niche events—that are not captured in existing Knowledge Bases (KBs). When Entity Linking (EL) systems fail to resolve these entities, the mentions are often discarded, creating gaps in downstream analysis. In this work, we study these long-tail entities using a large-scale corpus of one million news articles. To better understand their distribution, we extract entity mentions using a fine-grained taxonomy of 40 distinct types, and then attempt to link them to Wikipedia using an EL system.

1 Introduction

The news cycle operates at a pace that fundamentally outstrips the update cadence of curated Knowledge Bases (KBs) such as Wikidata and DBpedia. Each day, thousands of articles introduce entities — minor politicians, local businesses, regional cultural events, newly formed organizations — that have no corresponding KB entry. Named Entity Recognition (NER) and Entity Linking (EL) pipelines are the standard infrastructure for grounding natural language text in structured knowledge. Yet when EL fails, the prevailing practice is to silently discard the unresolvable mention, creating an invisible and unmeasured information loss.

This paper asks: how large is this gap, and where is it concentrated? We address this through a large-scale empirical study on the Signal-1M news corpus (Corney et al., 2016a), comprising one million English news articles spanning major wire services, regional outlets, and independent blogs. We adopt a unified 40-type entity schema combining OntoNotes 5.0 (Weischedel et al., 2013) and a targeted subset of FIGER (Ling and Weld, 2012) to capture fine-grained distinctions beyond standard coarse-grained taxonomies. For extraction, we use

GLiNER (Zaratiana et al., 2024), a prompt-based zero-shot NER model that accepts arbitrary type lists at inference time. For linking, we apply ReFinED (Ayoola et al., 2022b,a), a transformer that jointly performs fine-grained entity typing and disambiguation against Wikipedia.

Our main contributions are:

- A large-scale pipeline for fine-grained entity extraction and linking over one million news articles, yielding 5.4 million unique (text, type) pairs from 38.11 million raw extractions.
- Quantitative evidence that 57–76% of unique named entity mentions in news text cannot be linked to Wikipedia, with the gap concentrated in PRODUCT, EVENT, and MISCELLANEOUS types, and fine-grained rates ranging from 52% (*Politician*) to 97% (*Law*).
- A methodological comparison with Esquivel et al. (2017) showing that deduplication reduces apparent entity counts by 80–88% and that the long-tail phenomenon is a fundamental property of news-domain entity distributions, not an artefact of system choice.
- A manual error analysis of 450 linked entities establishing that NER–EL type agreement is a reliable correctness signal (100% precision on the diagonal) and identifying ORGANIZATION as the primary source of NER-side errors.

2 Related Work

2.1 The Knowledge Gap in News

A persistent challenge in news is that the entities most frequently discussed in local and niche media are precisely those least likely to appear in curated Knowledge Bases. The TAC Knowledge Base Population (KBP) shared tasks (Ji and Grishman, 2011) formalized this problem by defining NIL detection

and NIL clustering as explicit evaluation subtasks: systems must not only link mentions to a KB, but also identify mentions that have no corresponding entry and group them by real-world referent. While TAC KBP established the task definition, its benchmarks operated on relatively small, curated mention sets drawn from a narrow set of newswire sources.

At a structural level, Fetahu et al. (2015) demonstrated that Wikipedia systematically lags behind news events by days to months — new entities enter the news cycle long before any Wikipedia article is created for them, if one ever is. This temporal gap provides a direct structural explanation for the high no-overlap rates we observe in our study.

The closest direct predecessor to our work is Esquivel et al. (2017), who studied long-tail entities within Signal-1M using Stanford NER and DBpedia Spotlight across three coarse categories (PERSON, LOCATION, ORG). Their findings showed that person names are disproportionately difficult to link relative to other types. However, because their taxonomy is limited to three categories, it cannot reveal which finer-grained entity classes drive the gap — for instance, whether unlinkable entities are concentrated among minor public figures, local organizations, or cultural works. Our study addresses this directly by scaling to a 40-type schema and a more capable EL system, while remaining comparable to their baseline results for validation.

2.2 Entity Type Schemas

The choice of entity taxonomy determines the resolution at which the knowledge gap can be observed. OntoNotes 5.0 (Weischedel et al., 2013) provides 18 coarse-grained types and is the standard for general NER, but its broad categories — such as a single ORG type covering everything from multinational corporations to local sports clubs — mask the fine-grained distribution differences that matter for KB analysis. FIGER (Ling and Weld, 2012) addressed this by introducing a 2-level hierarchy of 112 types; however, Choi et al. (2018) demonstrated that very large type sets introduce substantial noise and reduce annotation reliability, particularly in sparse or zero-shot settings. More recently, MultiCoNER v2 (Fetahu et al., 2023) proposed a 33-class multilingual NER schema and highlighted the practical challenges of complex, noisy entities in real-world text. These studies collectively moti-

vate our choice of a 40-type schema, constructed by combining all 18 OntoNotes types with a targeted 22-type subset from FIGER. This positions our taxonomy between the coarse-grained standard and the noisy fine-grained extreme — expressive enough to reveal *what kind* of entity is absent from KBs, yet tractable enough to interpret reliably at scale.

2.3 NER and EL Systems

Named Entity Recognition. Early systems such as Stanford NER (Finkel et al., 2005) defined highly constrained type sets: a 3-class model (PERSON, ORGANIZATION, LOCATION), a 4-class extension adding MISC, and a 7-class variant covering temporal and numeric expressions. SpaCy (Honnibal et al., 2020) and Stanza (Qi et al., 2020) both adopt the OntoNotes 5.0 schema, offering broader coverage but remaining tied to its 18 fixed categories. NameTag 3 (Straková and Straka, 2025) supports multilingual NER but similarly falls back to OntoNotes or the 4-type CoNLL schema. All of these systems share a fundamental limitation for our purposes: their taxonomies are fixed at training time and cannot be extended without retraining.

GLiNER (Zaratiana et al., 2024) departs from this paradigm by framing NER as a span-labeling problem conditioned on a prompt-specified type list, abandoning the fixed-taxonomy constraint entirely. We adopt GLiNER because it allows us to inject our custom 40-type schema — including types absent from any single prior taxonomy — at inference time without retraining, making it the only practical choice for our combined OntoNotes-FIGER schema.

Entity Linking. DBpedia Spotlight (Mendes et al., 2011) links surface mentions to DBpedia using TF-IDF vectors and prior probabilities. Its primary strength is efficiency — it operates at high throughput with no neural inference cost, making it practical for large corpora. However, its reliance on surface-form matching makes it brittle against morphological variation and abbreviations, and its prior probability model heavily favors high-frequency entities, causing it to systematically underperform on rare or emerging entities — precisely the entities our study targets.

BLINK (Wu et al., 2020) substantially improves disambiguation quality by replacing sparse retrieval with a two-stage transformer architecture: a Bi-Encoder for fast candidate generation and a Cross-

Encoder for accurate re-ranking. This dense retrieval approach generalizes far better to rare entities and handles surface-form variation gracefully. Its main limitation, however, is that it treats entity linking as a pure disambiguation task — it assumes the mention is already known to be a named entity and produces no signal about whether the entity exists in the KB at all, outputting a best-guess Wikipedia entry regardless of confidence.

ReFinED (Ayoola et al., 2022b,a) addresses both limitations by jointly performing fine-grained entity typing and disambiguation against Wikipedia in a single transformer forward pass. By conditioning the linking decision on the predicted entity type, it achieves state-of-the-art accuracy on AIDA-CoNLL while remaining computationally feasible at scale. Crucially for our study, ReFinED explicitly models KB absence by producing a three-way outcome per mention — linked, NIL, or no overlap — which maps directly onto the long-tail signal we wish to measure. Its primary weakness is that it is anchored to Wikipedia as its sole KB; entities documented in Wikidata or domain-specific KBs but absent from Wikipedia are treated as un-linkable, which may cause our study to slightly overestimate the true knowledge gap. Nevertheless, we adopt ReFinED over alternatives precisely because its structured three-way output is the only design among available systems that makes KB incompleteness directly measurable at the scale of one million articles.

3 Method

3.1 Dataset

In this study, we use the Signal-1M dataset (Corney et al., 2016b), a publicly available corpus originally created for information retrieval research. It contains one million news articles collected over a one-month period in September 2015. The dataset includes content from a wide range of sources, such as major news outlets (e.g., Reuters, Associated Press), national and regional newspapers, blogs, and specialized publications.

Although most articles are in English, a small portion contains multilingual text. Each article includes structured metadata along with the main text, as shown in Figure 1. For entity extraction, we combine the `title` and `content` fields into a single input sequence.

Signal-1M Example Entry

id: 62c92c78-66e1-48e3-b6aa-2c3acb28854f
title: Emily Owsinski Named Principal Viola in Terre Haute Symphony Orchestra
source: The Peabody Post
media-type: Blog
published: 2015-09-21T20:34:11Z
content: Preparatory alumna Emily Owsinski, viola, has just been named Principal Viola in the Terre Haute Symphony Orchestra. The orchestra, now in its 90th season, is the oldest professional orchestra in the state of Indiana.

Figure 1: An example entry from the Signal-1M dataset.

Coarse Type	Fine Type
PERSON	PERSON, POLITICIAN, ATHLETE, MILITARY_PERSON, RELIGIOUS_LEADER
LOCATION	GPE, LOCATION, FACILITY
ORGANIZATION	ORGANIZATION, POLITICAL_PARTY, GOVERNMENT_AGENCY, SPORTS_TEAM, WEBSITE, NORP
EVENT	EVENT, ELECTION, MILITARY_CONFLICT, NATURAL_DISASTER, SPORTS_EVENT
PRODUCT	PRODUCT, WEAPON, VEHICLE, DISEASE, CHEMICAL_THING, LIVING_THING
WORK_OF_ART	WORK_OF_ART, FILM, MUSIC, WRITTEN_WORK
NUMERIC	DATE, TIME, PERCENT, MONEY, QUANTITY, ORDINAL, CARDINAL
MISCELLANEOUS	RELIGION, ETHNICITY, LANGUAGE, LAW

Table 1: The unified 40-type entity schema utilized in our pipeline, combining standard coarse-grained categories with fine-grained extensions from OntoNotes 5.0 and FIGER.

3.2 Entity Type Schema

To construct our entity extraction pipeline, we combine two taxonomies into a unified 40-type schema. The foundational component comprises the complete 18-type OntoNotes 5.0 schema (Weischedel et al., 2013). To better capture the nuanced, long-tail entities frequently found in news text, we augment this foundation with a 22-type subset derived from the FIGER taxonomy (Ling and Weld, 2012). These extended types were deliberately selected to introduce fine-grained distinctions within the broader PERSON, ORGANIZATION, and MISC categories. The complete taxonomy utilized in this study is detailed in Table 1.

3.3 Named Entity Recognition with GLiNER

We use GLiNER-large-v2.1 for entity extraction, prompting the model with the 40 fine-grained entity types defined in Table 1. The input to the model consists of the concatenation of each article’s title and content. Because GLiNER is constrained by a maximum context window of 380 tokens, we segment longer articles into processable chunks using the NLTK library.

Notably, GLiNER natively handles entity span resolution by prioritizing the longest valid match. For example, given the phrase “The University of Alberta,” the model correctly extracts the entire span as an ORGANIZATION, rather than incorrectly isolating the sub-span “Alberta” as a LOCATION. The output of this extraction stage is a set of (text, type) pairs representing the recognized entity mentions.

To avoid inflating our analyses with repeated mentions of the same entity surface forms—both within and across articles—we apply a strict deduplication step. We remove duplicate (text, type) pairs, retaining only the first occurrence of each unique combination.

3.4 Entity Linking with ReFinED

Entity linking is performed utilizing ReFinED, specifically employing the `wikipedia_model_with_numbers` checkpoint. ReFinED jointly performs mention detection, fine-grained entity typing, and entity disambiguation against Wikipedia within a single transformer forward pass. For each entity mention extracted by GLiNER, we pass the mention text along with its surrounding sentence context to ReFinED.

ReFinED assigns each mention one of three outcomes: (1) a mapped Wikipedia entity identifier (linked), (2) a NIL prediction, indicating the entity likely exists but has no corresponding KB, or (3) no overlap with the KB. We treat outcomes (2) and (3) collectively as the long-tail signal. However, we distinguish between them in our downstream analysis, as NIL mentions represent known unknowns, whereas no-overlap mentions fall entirely outside the scope of the KB.

ReFinED operates over standard semantic NER types from OntoNotes that are linkable to Wikipedia. Specifically, these represent the core entity categories: Person, Organization, GPE, Facility (FAC), Event, Work of Art, Product, and Lan-

guage. We explicitly exclude purely numeric types (Date, Time, Percent, Money, Quantity, Ordinal, and Cardinal), as they are not meaningful targets for entity linking.

4 Results

Running GLiNER on the full Signal-1M corpus produced 38.11 million entity mentions. After removing duplicates based on (text, type) pairs, this was reduced to 5.4 million unique entities. We evaluate EL outcomes on these entities. Each entity is assigned one of four outcomes: **Match**: ReFinED links the mention to a Wikipedia entity whose type matches the NER label at the coarse level. **Mismatch**: ReFinED links the mention to a Wikipedia entity, but the type does not match the NER label. **NIL**: ReFinED links the mention to a Wikipedia page, but no Wikidata type is available. **No Overlap**: ReFinED cannot link the mention to any Wikipedia entity.

4.1 Coarse-Level Entity Distribution

Table 2 reports the four EL outcomes aggregated by coarse category. Numeric types (DATE, MONEY, CARDINAL, etc.) are included for completeness but are excluded from further analysis.

The results reveal a clear stratification of linking outcomes. Location (23.83% match) and Person (22.49%) are the most linkable coarse categories, reflecting Wikipedia’s strong coverage of geographic entities and prominent individuals. Work of Art (18.58%) and Organization (13.22%) follow. At the other extreme, Product and Miscellaneous exhibit near-zero match rates (0.01% and 0.57% respectively), not because EL is inaccurate for these types, but because Wikipedia does not provide typed KB entries for the vast majority of commercial products, legal texts, and disease names appearing in news text.

The NIL column reveals a distinct failure mode from No Overlap. Miscellaneous has the highest NIL rate (24.09%) among named-entity categories, meaning nearly one in four Misc entities that Wikipedia *does* cover lacks a Wikidata type. Product follows at 21.58% NIL, while Person is lowest at 4.55%, consistent with Wikidata’s thorough typing of human entities.

4.2 Fine-Grained Breakdown by Category

Table 3 disaggregates the coarse results to the fine-grained NER types. Within each coarse group, sub-

Table 2: Entity linking outcomes by coarse NER category. *Match*: EL type agrees with NER coarse category. *Mismatch*: EL type disagrees. *NIL*: Linked to Wikipedia but Wikidata assigns no type. *No Overlap*: Cannot be linked to Wikipedia. Both NIL and No Overlap cases are treated as long-tail signals. Rows sorted by Match% descending.

Coarse Type	Total	Match (%)	Mismatch (%)	NIL (%)	No Overlap (%)
Location	654,449	23.83	6.64	12.21	57.32
Person	1,478,994	22.49	4.94	4.55	68.01
Work of Art	247,613	18.58	7.60	16.00	57.82
Organization	1,313,672	13.40	6.18	12.99	67.42
Event	316,419	4.34	4.37	15.44	75.85
Miscellaneous	157,669	0.57	5.52	24.09	69.81
Product	601,014	0.01	6.71	21.58	71.70
Numeric	790,599	0.00	0.96	3.76	95.28

types are sorted by Match% descending to surface the within-group variation.

We discuss the fine-grained results by category below.

Person. Within the Person category, *Politician* is the most linkable sub-type (40.12% match), reflecting Wikipedia’s systematic coverage of elected officials and public office holders. *Athlete* (28.59%) and *Religious Leader* (25.55%) follow, while the generic *Person* type is the least linkable (19.74%), a 20.38-point gap that illustrates Wikipedia’s selection bias toward prominence. Notably, the NIL rate for *Military Person* (14.16%) is substantially higher than for *Politician* (6.38%), suggesting that while military figures are often linkable, Wikidata’s typing of military entities is less complete.

Location. The generic *Location* type achieves the highest match rate in this group (28.42%), while *Facility* and *GPE* are substantially lower at 7.19% and 7.05% respectively. *GPE* (geopolitical entity) shows the highest NIL rate in the group (30.73%), suggesting that sub-national political units — boroughs, districts, and administrative divisions — are frequently linked to Wikipedia pages that Wikidata does not type as GPEs.

Organization. The Organization group shows the widest within-group variation, spanning from *Political Party* (27.35% match) to *Website* (3.98%), a 23.37-point range. *Sports Team* (18.34% match) stands out for its disproportionately high Mismatch rate (15.84%), the highest of any sub-type in the table. This occurs because many sports teams are disambiguated by ReFinED to their corresponding league or venue, which Wikidata types as an

organisation but not specifically as a sports team. *Ethnicity* (39.21% NIL) and *Religion* (38.26% NIL) show the two highest NIL rates across all fine-grained types. These entities are broadly covered by Wikipedia, but Wikidata’s type system does not provide unambiguous categories for ethnic groups or religions as institutions, resulting in typed pages that our mapping cannot resolve to the Organization coarse class.

Work of Art. Film achieves the highest match rate in this group (28.22%), followed by Music (19.36%) and Work of Art (16.40%). Written Work is the most long-tail sub-type (68.02% No Overlap), reflecting that most written works referenced in news — opinion columns, reports, press releases — are not individually documented in Wikipedia.

Event. The Event category has an Overall Match rate of only 4.34%, yet the Overall Mismatch rate is nearly identical (4.37%). This unusual pattern indicates that when an event entity *is* successfully linked, it is roughly as likely to receive a non-Event Wikidata type (typically Organisation or Location) as an Event type. *Sports Event* achieves the highest match rate in the group (12.40%), because major sporting events such as the FIFA World Cup are well-documented in Wikipedia with clear Wikidata event typing. *Military Conflict* has the highest NIL rate in the group (28.37%): battles and military operations are often linked to Wikipedia pages that Wikidata classifies as *conflict* or *military operation*, which our coarse mapping does not cover.

Miscellaneous. The Miscellaneous category encompasses types for which no EL column directly maps to the coarse class, explaining the near-zero

Match rates. *Disease* and *Law* have the highest No Overlap rates (75.04% and 74.20% respectively), driven by the use of informal, abbreviated, or colloquial names in news text that do not correspond to any Wikipedia page title. *Living Thing* shows the highest NIL rate in the group (31.58%), as many species and organisms are covered by Wikipedia but receive highly specific Wikidata biological classifications that our coarse mapping does not capture.

Product. The entire Product category has a Match rate of essentially zero (0.01%), as none of the EL output types correspond to a Product coarse class in our mapping. All linking mass falls in Mismatch, NIL, or No Overlap. *Weapon* and *Vehicle* have the highest NIL rates (30.77% and 28.72%) in the group, indicating that while individual weapons and vehicles are sometimes covered by Wikipedia, Wikidata’s typing of these entities is sparse. The generic *Product* sub-type has by far the largest No Overlap count in absolute terms (359,648 entities), making it the single largest contributor to the long tail across the entire dataset.

5 Discussion

5.1 Two Distinct Failure Modes

No Overlap and NIL represent qualitatively different problems with different remedies. **No Overlap** entities are absent from Wikipedia entirely and require new article creation. **NIL** entities exist in Wikipedia but lack a Wikidata P31 type and require structured editing of existing items. Conflating the two, as most prior work does, obscures which intervention is actually needed.

Disease (75.04% No Overlap) and *Law* (74.20%) are content gaps: the entities are simply not in Wikipedia. *Living Thing* (31.58% NIL), *Ethnicity* (39.21% NIL), and *Religion* (38.26% NIL) are typing gaps: Wikipedia covers them, but Wikidata’s type system is either too specific or too ambiguous to map cleanly to our coarse categories.

5.2 Comparison with Prior Work

Esquivel et al. (2017) analysed the same Signal-1M dataset using Stanford NER and DBpedia Spotlight. A key methodological difference is that their pipeline does not deduplicate entity mentions. To verify this, we reproduce their NER stage using Stanford NER on the same corpus and apply deduplication to the output. Table 4 shows that deduplication reduces their counts substantially — from

7.71M to 1.22M for PER, 5.52M to 0.69M for LOC, and 5.37M to 1.07M for ORG. Without deduplication, high-frequency, Wikipedia-notable entities dominate the statistics and artificially suppress the No Overlap rate.

We do not replicate their full EL pipeline. Instead, Table 5 places our deduplicated results alongside their published figures for direct comparison. Our No Overlap rates are consistently higher across all three types — most strikingly for LOCATION (57.32% vs. 7.33%) — confirming that deduplication exposes a long tail of rare, non-notable entities that frequency-weighted counting conceals. We also surface MISMATCH as a separate outcome, showing that a non-trivial fraction of apparently successful links receive the wrong Wikidata type, a failure mode invisible in prior work.

5.3 Error Analysis

To validate our result, we manually annotated a stratified sample of 450 linked entities — 50 per cell of a 3×3 grid crossing coarse NER types (PERSON, ORGANIZATION, LOCATION) with EL-assigned types — labelled *Both Correct*, *NER Correct* (NER right, EL entity wrong), or *EL Correct* (EL entity right, NER label wrong). Results are in Table 6.

All three agreement cells (PERSON→PERSON, ORG→ORG, LOC→LOC) achieve 100% *Both Correct*, confirming that NER–EL type agreement is a reliable correctness signal. In disagreement cells, error responsibility is asymmetric. GLiNER’s PERSON is reliable (78–98% *NER Correct*), while ORGANIZATION is not: 84–86% of ORG→PERSON and ORG→LOC cases are *EL Correct*, meaning GLiNER mislabelled persons and facilities as organisations. The LOC→PERSON cell is evenly split, reflecting surface-form ambiguity between surnames and place names.

5.4 Limitations

We note three limitations.

First, the Signal-1M dataset dates to September 2015, and both Wikipedia and Wikidata have expanded significantly since then. As a result, our No Overlap rates likely represent an upper bound compared to what would be observed on a more recent corpus. Replicating this analysis on a contemporary dataset such as the NELA-PS news dataset (Horne and Gruppi, 2024) would provide a more up-to-date estimate.

Second, our coarse mapping from Wikidata types to NER categories is not perfect, especially for complex classes such as *Military Conflict* and *Living Thing*. These categories have rich hierarchies that cannot be easily reduced to a single label, so some Mismatch and NIL cases may be caused by the mapping rather than true EL errors.

Third, our error analysis (Section 5.3) is limited in scope, covering only three coarse types and 450 entities. Expanding this analysis to all 40 entity types would provide more reliable and detailed error estimates.

6 Conclusion

We presented a large-scale study of long-tail entities in one million English news articles from the Signal-1M corpus. Running GLiNER with a 30-type schema over the full corpus yielded 38.11 million entity mentions; after deduplication on (text, type) pairs, this reduced to 5.4 million unique entities, which we linked against Wikipedia using ReFinED.¹

Our central finding is that 57–76% of unique named entity mentions in news text cannot be usefully grounded in a KB, and that this gap is structured rather than uniform. Product, Event, and Miscellaneous entities are almost entirely unresolvable, while Person and Location entities fare comparatively better due to Wikipedia’s coverage of prominent individuals and places. Within categories, fine-grained types reveal stark variation: *Politician* entities are linked at more than twice the rate of the generic *Person* type, and *Film* is linked at four times the rate of *Written Work*.

We showed that the long tail comprises two qualitatively different failure modes — No Overlap (absent from Wikipedia) and NIL (present in Wikipedia but untyped in Wikidata) — that require different remedies and should not be conflated. Comparison with Esquivel et al. (2017) on the same corpus revealed that deduplication is a critical methodological decision: without it, high-frequency notable entities dominate the statistics and mask the true extent of the gap. Finally, our manual error analysis of 450 entities confirmed that NER–EL type agreement is a reliable correctness signal with 100% precision on the diagonal, and identified GLiNER’s ORGANIZATION predictions

as the primary source of NER-side errors, driven by person-named commercial entities and facility names.

These findings establish a quantitative baseline for the entity long tail in news text and provide actionable targets for KB population: PRODUCT and EVENT entities for new article creation, and *Ethnicity*, *Religion*, and *Living Thing* entities for Wikidata typing campaigns.

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¹Code and data are available at <https://github.com/punreachrany/cmp656-long-tail-entities-news.git>.

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Table 3: Entity linking outcomes by fine-grained NER type, grouped by coarse category (bold rows). Both NIL and No Overlap cases are treated as long-tail signals. Sub-types sorted by Match% descending within each group.

NER Type	Total	Match (%)	Mismatch (%)	NIL (%)	No Overlap (%)
PERSON					
Politician	103,171	40.12	7.56	6.38	45.94
Athlete	207,565	28.59	5.09	3.75	62.57
Religious Leader	21,616	25.55	8.56	11.94	53.96
Military Person	47,340	19.95	9.87	14.16	56.02
Person	1,099,302	19.74	4.39	3.97	71.91
Overall	1,478,994	22.49	4.94	4.55	68.01
LOCATION					
Location	512,944	28.42	6.71	11.30	53.56
Facility	137,408	7.19	6.33	15.03	71.45
GPE	4,097	7.05	8.93	30.73	53.28
Overall	654,449	23.83	6.64	12.21	57.32
ORGANIZATION					
Political Party	17,163	27.35	4.52	21.00	47.12
Government Agency	61,342	26.63	3.61	18.02	51.74
Sports Team	74,239	18.34	15.84	16.01	49.82
Organization	977,295	13.72	6.17	13.43	66.67
Website	182,250	3.98	3.23	6.85	85.94
Norp	1,383	3.62	20.75	25.89	49.75
Overall	1,313,672	13.40	6.19	12.99	67.42
WORK OF ART					
Film	48,974	28.22	9.76	17.22	44.80
Music	92,005	19.36	7.87	15.06	57.72
Work of Art	45,818	16.40	6.48	18.71	58.41
Written Work	60,816	11.27	6.29	14.42	68.02
Overall	247,613	18.58	7.60	16.00	57.82
EVENT					
Sports Event	50,940	12.40	4.39	14.55	68.67
Election	7,580	4.55	2.57	16.28	76.60
Event	219,612	3.04	4.15	13.80	79.01
Natural Disaster	16,451	1.42	6.19	22.59	69.80
Military Conflict	21,836	0.72	5.72	28.37	65.18
Overall	316,419	4.34	4.37	15.44	75.85
MISCELLANEOUS					
Religion	8,192	14.51	4.79	38.26	42.44
Language	11,157	14.31	3.63	25.37	56.69
Ethnicity	16,699	11.68	5.01	39.21	44.10
Living Thing	26,095	5.99	3.89	31.58	58.54
Disease	25,543	2.16	1.97	20.82	75.04
Law	69,983	2.02	1.48	22.29	74.20
Overall	157,669	5.24	2.66	26.43	65.67
PRODUCT					
Product	489,462	0.008	6.45	20.07	73.48
Vehicle	53,504	0.003	8.45	28.72	62.83
Chemical Thing	38,674	0.00	7.28	26.28	66.44
Weapon	19,374	0.00	7.41	30.77	61.82
Overall	601,014	0.006	6.71	21.58	71.70

Table 4: Entity counts before and after deduplication.

	Previous Work		This work	
	All	Unique	All	Unique
PER	7.71M	1.22M	8.92M	1.48M
LOC	5.52M	0.69M	6.07M	0.65M
ORG	5.37M	1.07M	10.29M	1.31M

Table 5: Distribution of EL outcomes across NER types. We compare prior work [Esquivel et al. \(2017\)](#) (Stanford NER + DBpedia Spotlight) with our approach (GLiNER + ReFinED). Both NIL and No Overlap cases are treated as long-tail signals.

NER Type	PER	LOC	ORG	MISC	NIL	No Overlap
<i>Previous work</i>						
PER (7.71M)	26.59%	4.71%	2.52%	1.29%	10.51%	54.38%
LOC (5.52M)	0.65%	64.55%	6.43%	1.62%	19.42%	7.33%
ORG (5.37M)	1.49%	11.91%	39.44%	4.68%	27.94%	14.54%
<i>This work</i>						
PER (1.48M)	22.49%	3.54%	0.88%	0.53%	4.55%	68.01%
LOC (0.65M)	0.65%	23.83%	5.02%	0.97%	12.21%	57.32%
ORG (1.34M)	0.53%	3.96%	13.40%	1.70%	12.99%	67.42%

Table 6: Manual annotation of 450 linked entities (50 per NER×EL cell). *Both*: both correct. *NER*: NER correct, EL entity wrong. *EL*: EL entity correct, NER label wrong.

NER	EL Type	Both (%)	NER (%)	EL (%)
PERSON	PERSON	100	0	0
	ORG	10	78	12
	LOC	0	98	2
ORG	PERSON	0	14	86
	ORG	100	0	0
	LOC	0	16	84
LOC	PERSON	2	48	50
	ORG	0	100	0
	LOC	100	0	0